

Source, Forge, Deploy, Nexus

A Cognitive Lifecycle Architecture for Proof-Bounded NeuroAI Systems

Series: *Repo-Native NeuroAI Infrastructure*

Paper: II of IV

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Authority state: CLAIM_HARDENED / PUBLIC_SAFE_DRAFT

Primary substrate: R2, R3, R4, R5, R6, R7 with action/perception context from E5, E6, E7

Claim posture: repo-supported protocol architecture plus C4 strategic thesis

Boundary: no clinical validation, biological equivalence, production deployment, external notarization, legal filing, or CERN affiliation is claimed.

Next proof move: build trace-based artifact cognition tasks that measure source fidelity, forged-claim safety, deployment gating, and nexus registration.

Abstract

Intelligence is not only what an agent thinks. It is how raw material becomes authorized action without losing provenance. Medina's internal repo intelligence defines four active surfaces: Source, Forge, Deploy, and Nexus. Source interprets doctrine and substrate. Forge turns source material into artifacts. Deploy evaluates release, promotion, runtime implications, and verification gates. Nexus registers stable outputs as lineage objects, manifests, schemas, and shared contracts. This paper frames that four-surface architecture as a cognitive lifecycle model for proof-bounded NeuroAI systems. The claim is not that Source/Forge/Deploy/Nexus is active inference, a biological circuit, or a deployed external service. The claim is that it supplies a repo-native model of artificial cognition as regulated transformation: perception, synthesis, action-readiness, and durable registration under explicit claim boundaries.

1. The Four Verbs of Cognitive Infrastructure

An agent does not become intelligent merely by producing text. A research agent must transform source into consequence. It must see what kind of material it has encountered, decide what can be made from it, test whether the result may move into the world, and preserve the result so that future work can inherit it without confusion. That sequence is not a convenience. It is a cognitive lifecycle.

Medina's repo intelligence names the lifecycle with unusual clarity: **Source, Forge, Deploy, Nexus**. These are not just folders or workflow labels. The internal repo map says to treat them as active compartments. Source receives doctrine, architecture, protocol intelligence, and high-authority patterns. Forge receives drafts, ledgers, schemas, modules, and generated artifacts. Deploy receives release logic, promotion checks, deployment implications, and readiness gates. Nexus receives stable contracts, shared manifests, registries, and lineage objects.

The sequence is simple enough to remember and deep enough to test:

Source sees. Forge forms. Deploy gates. Nexus remembers.

That line is the technical and narrative core of this paper.

2. From Agent Loop to Research Lifecycle

AI agents are often described through loops: perceive, plan, act; observe, think, execute; retrieve, reason, respond. Those loops are useful but under-specified for research work. They do not say what counts as source truth. They do not say when synthesis becomes a claim. They do not say how a draft becomes public-safe. They do not say how a result becomes durable memory rather than orphaned output.

The Source/Forge/Deploy/Nexus model is more severe. It treats each transition as a promotion event with boundary conditions.

Stage	Cognitive Verb	Artifact Question	Failure Mode
Source	perceive / classify	What is this material, and what authority does it carry?	source hallucination or authority confusion
Forge	synthesize / embody	What artifact can be built without overclaiming?	forged overclaim or unsupported theorem
Deploy	gate / act	Where may the artifact go, and what proof is required?	unsafe release or deployment fiction
Nexus	register / remember	What becomes stable lineage for future work?	orphaned artifact or false canon

This model matters because the failure modes are not hypothetical. A system can summarize private doctrine into public prose. It can turn a template into an invented implementation claim. It can generate a deployment plan and imply that deployment exists. It can create a file and lose the lineage that explains why it matters. These are not style errors. They are cognitive lifecycle failures.

3. Source Is Perception

Source is not ingestion. It is structured perception. The internal repo’s Source-to-Forge flow begins by extracting doctrine, source logic, and high-authority patterns, then classifying substrate and release boundary before converting material into artifacts. This is perception with epistemic depth: the agent must see not only words but authority.

For Medina-native NeuroAI, Source perception includes at least six judgments:

1. substrate type: doctrine, code, ledger, template, proof record, runtime note, or public claim;
2. authority class: trunk law, candidate, internal signal, verified proof, unsupported claim, or private material;
3. source identity: user-provided, repo-derived, attached, external, or generated;
4. proof posture: verified, supported, weak, asserted, mixed, or private;
5. release boundary: public-safe, internal-only, IP-review-first, private-vault, or evidence-needed;
6. downstream route: paper, packet, proof ledger, deployment bridge, archive, or notary prep.

A generic agent may read text. Source sees a claim’s future consequences.

Claim Boundary Box 2

Supported: Source-to-Forge doctrine exists in repo files.

Hypothesis: source classification functions as an artificial perception layer for research cognition.

Missing proof: benchmarked comparisons between source-aware and source-blind agents.

Blocked claim: Source is a biological perceptual system.

4. Forge Is Where Claims Become Bodies

Forge is the stage where material becomes artifact. It is tempting to treat this as writing or generation, but the repo’s Forge logic is broader. Forge can produce structured markdown papers, claims matrices, evidence matrices, proof ledgers, schema drafts, code-generation plans, packet builders, and branchable artifacts. It is where thought acquires body.

The danger of Forge is overformation. A weak source can be forged into a confident paper. A candidate theorem can be dressed as proof. A private architecture can become a public diagram. A placeholder can become a product claim. Forge is powerful because it makes things real enough to circulate. That is exactly why it must inherit Source’s authority labels.

In a Medina NeuroAI context, Forge should not be judged only by beauty or completeness. It should be judged by transformation fidelity. Did it preserve evidence state? Did it mark hypotheses? Did it keep private mechanisms abstract? Did it avoid collapsing a packet stack into a single promotional artifact? Did it generate the matrices and manifests needed for later review?

The best forged artifact is not the loudest document. It is the one whose structure knows what it is allowed to say.

5. Deploy Is a Gate, Not a Button

Deploy is often treated as a final act: ship the code, publish the paper, release the model. Medina's repo intelligence treats Deploy more carefully. The Forge-to-Deploy rail says generated work should be reviewed for claim and evidence integrity, release boundaries, protected details, deployment implications, runtime placement, rollback needs, hash-manifest path, and notary-prep status.

In other words, deployment is not motion. It is judgment.

For NeuroAI agents, this is crucial. An agent that can generate an artifact must also know where that artifact may go. Internal research can contain private architecture. A public-safe derivative must abstract it. A deployment bridge needs environment boundaries and rollback plans. A notary packet needs hash manifests but cannot claim external witnessing without a receipt.

Deploy therefore resembles action selection under inhibition. It is not enough to know what action is possible. The system must know what action is permitted.

This is where Medina's architecture connects to active inference and action-perception theories only by analogy. Predictive coding and active inference treat perception and action as coupled under generative models, but reviews warn that empirical support and model comparison remain active concerns. Source/Forge/Deploy/Nexus should not be called active inference. It should be compared as a software architecture hypothesis: a bounded action loop for artifact-producing cognition.

6. Nexus Prevents Orphaned Intelligence

Nexus is where output becomes durable enough to matter. The Nexus Registration Flow registers stable outputs as manifests, schemas, contracts, lineage objects, and interoperability references. The rule is equally important: draft-only or unstable artifacts should stay out of the shared registry layer.

This is memory with admission control. Not everything generated deserves to become canon. Not everything useful is stable. Not every draft should become a contract. Nexus is the difference between a pile of artifacts and an evolving intelligence system with continuity.

The repo's Agent Engine Manifest strengthens the point. Stable engine roles, packet schemas, manifest schemas, release-boundary contracts, deployment translation contracts, and lineage objects can become reusable shared contracts. That is how a one-off result becomes part of Medina's future cognition.

7. Anti-Fragmentation and the Heart of the Loop

The Heavy and Weights Heart file adds the missing emotional-intellectual discipline: preserve one organism-level result even when multiple internal roles are active; prefer deep coherence over superficial completion; do not let multi-render output create contradictory authority states; keep internal, external, archive, and deployment variants aligned to one claim and evidence backbone.

This is more than workflow hygiene. It is an anti-fragmentation principle. Long agent runs split easily. One surface drafts, another packages, another summarizes, another deploys. Without a coherence rule, the system can produce beautiful fragments that contradict each other.

A serious NeuroAI program should treat anti-fragmentation as a measurable property. Can an agent preserve one authority model while generating a paper, public summary, claims matrix, evidence matrix, hash manifest, and notary packet? Can it keep the public version abstract while the internal version remains complete? Can it change a claim in one artifact and propagate the boundary through the packet?

8. The Lifecycle Has Failure Modes

The four-stage loop allows precise failures:

Failure	Stage	Description	Repair
Source hallucination	Source	agent invents source authority or misreads substrate	require source identity and authority class
Forge overclaim	Forge	draft promotes weak claim into proof	claims matrix and citation hardening
Deploy leakage	Deploy	public artifact exposes internal/private material	release-boundary gate
Nexus orphaning	Nexus	artifact exists without lineage or promotion state	lineage record and manifest registration
Lifecycle collapse	all	agent jumps from raw source to public release	packet pipeline enforcement

These failures are testable. That is why the lifecycle belongs in a research paper rather than an internal instruction sheet.

9. Experimental Program

A benchmark for Source/Forge/Deploy/Nexus should provide mixed repo surfaces and ask agents to produce a complete packet. The evaluation should include source classification accuracy, transformation fidelity, release-gate correctness, lineage completeness, contradiction avoidance, and recovery after correction.

One possible task is the **Messy Repo-to-Packet Challenge**. The input contains canonical docs, outdated drafts, private notes, templates, proof ledgers, and misleading claims. The agent must produce a public-safe preprint packet. Scoring penalizes invented evidence, private leakage, orphaned artifacts, missing hash manifests, and failure to downgrade unsupported claims.

This is not a benchmark of politeness or prose. It is a benchmark of artifact cognition.

10. Objections

Objection 1: The four surfaces are just software workflow.

Response: yes, they are software workflow. The research claim is that workflow becomes cognitive when the system must preserve authority, proof, memory, and action boundaries across time.

Objection 2: The lifecycle may slow creative work.

Response: Source/Forge/Deploy/Nexus should not govern every scratch note. It should govern consequential artifacts: public papers, IP packets, deployment plans, proof ledgers, and notary-prep bundles.

Objection 3: The model is too Medina-specific.

Response: the naming is Medina-specific; the general pattern is not. Many research agents will need perception of source authority, controlled synthesis, release gating, and memory registration.

11. Conclusion: Intelligence Is a Path, Not a Moment

Source, Forge, Deploy, and Nexus turn intelligence from a moment of output into a path of transformation. Medina’s repo intelligence shows how raw material can become artifact, artifact can become action candidate, and action candidate can become lineage. The NeuroAI contribution is not that this loop is biological. It is that cognitive systems worth trusting must transform evidence without losing the boundaries that make knowledge safe.

12. Formalizing the Lifecycle

The Source/Forge/Deploy/Nexus loop can be formalized as a typed transition system. Let **S** be a source object with fields for identity, authority, substrate class, and evidence state. Let **F** be a forged artifact with fields for claim class, evidence mapping, release boundary, and derivative policy. Let **D** be a deployment or release decision with

fields for destination, proof requirement, rollback condition, and blocked language. Let **N** be a nexus registration object with fields for lineage, manifest reference, review state, and promotion state.

The lifecycle is safe only if transitions preserve required fields:

- **Source** -> **Forge** must preserve source identity and authority state.
- **Forge** -> **Deploy** must preserve claim class, evidence posture, and release boundary.
- **Deploy** -> **Nexus** must preserve review decision, destination surface, and lineage reference.
- **Nexus** -> **Source** must allow future runs to retrieve the registered object without confusing it for stronger proof than it carries.

This structure creates a measurable invariant: no artifact may gain authority merely by moving stages. Authority can increase only when evidence or review state changes. A draft does not become public-safe because it was packaged. A local hash does not become notarization because it was registered. A deployment plan does not become deployment because it entered a Deploy surface.

13. Lifecycle Invariants

The following invariants should hold for a proof-bounded cognitive lifecycle:

1. **Source conservation:** every major artifact can name the source objects that gave rise to it.
2. **Claim conservation:** no claim becomes stronger without explicit evidence or review.
3. **Boundary conservation:** private and public variants remain linked to one release policy.
4. **Lineage conservation:** every stable output can be traced to a packet, revision, and manifest.
5. **Promotion conservation:** generated form does not imply stronger authority.

These invariants make the lifecycle catchy but not casual. The loop is not merely “workflow.” It is a discipline for artificial cognition under transformation pressure.

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Repo Intelligence Substrate

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- R2. `/workspace/agent_files/internal-repo/mini-brain.md` — THESIS routing scaffold for multi-surface work.
- R3. `/workspace/agent_files/internal-repo/heavy-and-weights-heart.md` — coherence, continuity, and anti-fragmentation modulation reference.
- R4. `/workspace/agent_files/internal-repo/source-to-forge.md` — doctrine-to-artifact transformation rail.
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